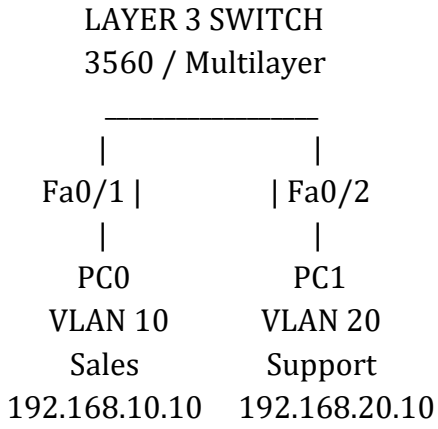
DAY_6

- **Layer 3 Switch Inter-VLAN Routing???? Theory: Using a Layer 3 SwitchNo need for a routerUses SVIs (Switch Virtual Interfaces) for routingFaster than Router-on-a-Stick???? Practical: Configuring Inter-VLAN Routing on a Layer 3 Switch???? Hands-on LabEnable IP Routing**

1. On a Cisco Layer 3 switch, create three VLANs (e.g., VLAN 10 for 'Sales', VLAN 20 for 'Support', VLAN 30 for 'Marketing') and assign at least one interface to each VLAN.

ANS :

- **What Is Layer 3 Switch Inter-VLAN Routing?**
  - A normal **Layer 2 switch** can switch traffic inside the same VLAN, but devices in different VLANs cannot communicate directly.



- **IP Addressing Table**

Device	VLAN	IP Address	Subnet Mask	Default Gateway
PC0	10 – Sales	192.168.10.10	255.255.255.0	192.168.10.1
PC1	20 – Support	192.168.20.10	255.255.255.0	192.168.20.1
PC2	30 – Marketing	192.168.30.10	255.255.255.0*	192.168.30.1

2. Configure Switch Virtual Interfaces (SVIs) for each VLAN you created by assigning them unique IP addresses (e.g., 192.168.10.1/24 for VLAN 10, etc.) and ensure each SVI is in 'up/up' state. Hint: Use the 'interface vlan' command to create SVIs and 'no shutdown' to enable them.

ANS :

- **VLAN 10 – Sales SVI**

```
configure terminal

interface vlan 10
ip address 192.168.10.1 255.255.255.0
no shutdown
    exit
```

- **VLAN 20 – Support SVI**

```
interface vlan 20
ip address 192.168.20.1 255.255.255.0
no shutdown
    exit
```

- **VLAN 30 – Marketing SVI**

```
interface vlan 30
ip address 192.168.30.1 255.255.255.0
no shutdown
    exit
```

3. Enable IP routing on the Layer 3 switch so that devices in different VLANs can communicate with each other without using an external router. Hint: Use the 'ip routing' global configuration command.

ANS :

- **Enable IP Routing**

  Enable routing:
  configure terminal
  ip routing
  end
  This is the most important command for Layer 3 inter-VLAN routing:
  ip routing
  The switch can now route between:
  192.168.10.0/24
  ↕
  192.168.20.0/24
  ↕
  192.168.30.0/24
  No external router is required.

- **Configure the PCs**

  PC0 – VLAN 10 / Sales
  Go to:
  Desktop → IP Configuration
  IP Address: 192.168.10.10
  Subnet Mask: 255.255.255.0
  Default Gateway: 192.168.10.1

  PC1 – VLAN 20 / Support
  IP Address: 192.168.20.10
  Subnet Mask: 255.255.255.0
  Default Gateway: 192.168.20.1

  PC2 – VLAN 30 / Marketing
  Initially:
  IP Address: 192.168.30.10
  Subnet Mask: 255.255.255.0
  Default Gateway: 192.168.30.1

4. Verify inter-VLAN routing by connecting two PCs (one in VLAN 10 and one in VLAN 20) and pinging between them. Capture and submit the ping results to show successful communication.

ANS :

- **Verify Inter-VLAN Routing**

From **PC0 (Sales – VLAN 10)**, open:
Desktop → Command Prompt
Run:
ping 192.168.20.10
Expected result:
Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
Why Does the Ping Succeed?
Traffic travels like this:
PC0
192.168.10.10
VLAN 10
|
| Default Gateway
v
SVI VLAN 10
192.168.10.1
|
| Layer 3 Switch Routing
v
SVI VLAN 20
192.168.20.1
|
v
PC1
192.168.20.10
VLAN 20

The ping succeeds because:

1. PC0 and PC1 have correct IP addresses.
2. Their default gateways point to the Layer 3 switch SVIs.
3. VLAN 10 and VLAN 20 are active.
4. The correct ports are assigned to their VLANs.
5. The SVIs are configured with IP addresses.
6. ip routing is enabled.

5. Refactor your VLAN and SVI configuration so that VLAN 30 uses a different subnet mask (e.g., /25 instead of /24). Explain what changes you made and how this affects communication with the other VLANs.

ANS :

- **Change VLAN 30 to /25**

Now change the Marketing VLAN subnet mask from:
Old: 192.168.30.0/24
Mask: 255.255.255.0
to:
New: 192.168.30.0/25
Mask: 255.255.255.128
What Is /25?
A /25 subnet mask is:
255.255.255.128

Subnet	Usable Host Range	Broadcast
192.168.30.0/25	192.168.30.1 – 192.168.30.126	192.168.30.127
192.168.30.128/25	192.168.30.129 – 192.168.30.254	192.168.30.255

- **Complete Layer 3 Switch Configuration**

```
enable
configure terminal

! ==========================
! CREATE VLANs
! ==========================

vlan 10
name Sales
exit

vlan 20
name Support
exit

vlan 30
name Marketing
exit

! ==========================
! ASSIGN ACCESS PORTS
! ==========================

interface fastEthernet0/1
switchport mode access
switchport access vlan 10
no shutdown
exit

interface fastEthernet0/2
switchport mode access
switchport access vlan 20
no shutdown
exit

interface fastEthernet0/3
switchport mode access
switchport access vlan 30
no shutdown
exit

! ==========================
! CONFIGURE VLAN 10 SVI
! ==========================
```

```
interface vlan 10
ip address 192.168.10.1 255.255.255.0
no shutdown
exit

! ==========================
! CONFIGURE VLAN 20 SVI
! ==========================

interface vlan 20
ip address 192.168.20.1 255.255.255.0
no shutdown
exit

! ==========================
! CONFIGURE VLAN 30 SVI
! Using /25
! ==========================

interface vlan 30
ip address 192.168.30.1 255.255.255.128
no shutdown
exit

! ==========================
! ENABLE LAYER 3 ROUTING
! ==========================

ip routing

end
    write memory
```